Session 7: Experimental Design & Analysis of Variance (ANOVA)

Context & Objective

Textbook: Chapter 13 – Experimental Design & Analysis of Variance

Objective: Students evaluate a pricing experiment across three regions using ANOVA and Fisher's LSD to choose the optimum pricing strategy.

1. Antigravity Skill Definition (anova-variance-analyzer)

Skill Name: anova-variance-analyzer

Purpose: Performs One-way ANOVA for completely randomized experimental designs. Computes F-statistic, p-value, and Fisher's LSD for pairwise comparisons to test if group means differ significantly.

1.1 Core Logic & Formulas

Step 1: Input Data Arrays for k Groups

Organize data from k groups (pricing tiers in this case).

Step 2: Compute Grand Mean, SSTR, SSE, and SST

Grand Mean: Average of all observations across all groups

SSTR (Sum of Squares Treatment): $\Sigma\, n_i(\bar{x}_i - \bar{x})^2$ – variation between groups

SSE (Sum of Squares Error): $\Sigma\,\Sigma\, (x_{ij} - \bar{x}_i)^2$ – variation within groups

SST (Total Sum of Squares): SSTR + SSE – total variation

Step 3: Compute Mean Squares and F-Statistic

MSTR = SSTR / (k - 1) where k = number of groups

MSE = SSE / (n_t - k) where n_t = total number of observations

F-statistic = MSTR / MSE with degrees of freedom (k-1, n_t-k)

Step 4: Decision Rule

If F > F_α(k-1, n_t-k) or p-value < α: Reject H_0 (significant difference exists)

If F ≤ F_α(k-1, n_t-k) or p-value ≥ α: Fail to Reject H_0 (no significant difference)

Step 5: Post-Hoc Pairwise Comparisons (if p-value < α)

Calculate Fisher's LSD:

$$LSD = t_{\alpha/2,\, n_t-k} \times \sqrt{[\, MSE \times (1/n_i + 1/n_j)\,]}$$

For each pair (i, j): If $|\bar{x}_i - \bar{x}_j| > LSD \rightarrow$ significant difference

2. Student Execution Prompt

Compare sales volumes under three pricing tiers (Low, Medium, High) tested in 4 stores each:

Low Pricing Tier: [500, 520, 480, 510]

Medium Pricing Tier: [460, 480, 450, 470]

High Pricing Tier: [380, 400, 370, 390]

Task: Perform a One-way ANOVA at significance level α = 0.05. Calculate the F-statistic, degrees of freedom, and p-value. Then compare mean differences using Fisher's LSD (t_critical = 2.228 for this design).

Summary Statistics by Pricing Tier:

Pricing Tier	n	Mean ($\bar{x}$)	Std Dev (s)
Low	4	502.5	15.43
Medium	4	465.0	13.23
High	4	385.0	11.18

3. Managerial Decision Output

ANOVA Results & Interpretation:

F-Statistic: 98.4
p-Value: < 0.001
Fisher's LSD: 19.3

Pairwise Comparisons:

Low vs Medium: Difference = 37.5 (> 19.3) → Significant
Low vs High: Difference = 117.5 (> 19.3) → Significant
Medium vs High: Difference = 80.0 (> 19.3) → Significant

Managerial Decision:

The sales manager's analysis reveals highly significant differences across pricing tiers (F = 98.4, p < 0.001). All pairwise comparisons exceed Fisher's LSD threshold of 19.3, confirming that each pricing strategy performs distinctly. The Low pricing tier generates mean sales of 502.5 units per store, substantially outperforming both Medium (465.0) and High (385.0) tiers. Despite the lower unit margin associated with the Low price point, the volume gains are statistically significant and economically compelling. The manager decides to implement the Low pricing tier across all regions, predicting this strategy will maximize total revenue and market share while maintaining operational efficiency.

